

Serial:

ID:

Department of Mathematical and Physical Sciences, EWU
Final Examination, Spring-2026
Course: STA 102 (Statistics and Probability), S-10

Full Marks: 30, Time: 1 Hour & 15 Minutes

Answer all the questions. Figures in the right margin indicate marks.

1. a) John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5. What is the probability that he will get at least one offer from these two companies? (2)
- b) In a university, consider the following events for students in the CSE and Data Science program: (4)
- C : A student has strong programming skills
 D : A student is proficient in data analysis
- The probabilities are given as:
 $P(C) = 0.75, P(D) = 0.70, P(C \cup D) = 0.90$
- (i) What is the probability that a student is proficient in data analysis given that they have strong programming skills?
- (ii) What is the probability that a student has strong programming skills given that they are proficient in data analysis?
- (iii) Interpret the results: Which skill is more conditionally dependent on the other?
- c) A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently. (2)
- d) A tech company employs two data scientists to build predictive models. Data Scientist 1 develops models for 70% of the projects, while Data Scientist 2 develops models for the remaining 30%. It is known that the probability that Data Scientist 1 produces a model with a critical error is 0.02, whereas Data Scientist 2 produces a model with a critical error is 0.04. Now suppose a deployed model is found to contain a serious error. (i) What is the probability that Data Scientist 1 developed the faulty model? (ii) Based on your result, which data scientist would you most likely suspect? Justify your answer. (4)
2. a) In an automated textile manufacturing system, computer vision algorithms are used to detect imperfections in fabric rolls. Let X denote the number of detected imperfections per 10 meters of synthetic fabric. The probability distribution of X is given below: (2)
- | | | | | | |
|--------|------|------|------|------|------|
| x | 0 | 1 | 2 | 3 | 4 |
| $p(x)$ | 0.41 | 0.37 | 0.16 | 0.05 | 0.01 |
- (i) Compute the expected number of imperfections per 10 meters of fabric.
- (ii) Compute the standard deviation of the number of imperfections.

- b) In many data-driven systems, such as sensor networks and scientific computing applications, measurements are subject to random error. Accurate modeling of these errors is essential for reliable system performance. Suppose the measurement error X of a certain system follows a probability density function given by: (4)

$$f(x) = \begin{cases} 0.1875(3 - x^2), & -1 \leq x \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

- (i) Find the cumulative distribution function (CDF).
 - (ii) Find the probability that the measurement error is less than 0.5.
 - (iii) For this system, it is considered undesirable if the *magnitude of the error exceeds 0.8*, i.e., $X < -0.8$ or $X > 0.8$. Compute the probability that this occurs.
3. a) A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%. The inspector randomly picks 20 items from a shipment. What is the probability that there will be at most one defective item among these 20? (2)
- b) In a cloud computing system, incoming user requests to a server follow a Poisson process with an average rate of 2.5 requests per minute. Let X denote the number of requests received in a given one-minute interval. What is the probability that at least 2 requests are received in a minute? (2)
- c) In a cloud-based application, the response time (latency) of a server is approximately normally distributed with a mean of 200 milliseconds and a standard deviation of 15 milliseconds. Let X denote the response time (in milliseconds) for a randomly selected request. (3)
- (i) What is the probability that the response time exceeds 240 ms?
 - (ii) Determine the response time below which 95% of requests are completed.
4. A data scientist analyzes the average response time of an API in a cloud service. Historically, the mean response time is believed to be 250 ms. To evaluate current performance, a random sample of 50 API requests is collected. The sample yields a mean response time of 240 ms with a standard deviation of 30 ms. (5)
- (i) Construct a 95% confidence interval for the true mean response time of the API.
 - (ii) The data scientist also wants to test whether the current average response time has improved (i.e., decreased) from the historical value of 250 ms. Compute the p-value for the hypothesis test and state your conclusion.